

DAY-8

DATABASE SETUP

1. Database Setup

- Database setup means configuring and preparing a database so that an application can store and retrieve data.
 - Examples of databases:
 - **MongoDB**
 - **MySQL**
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2. Virtual Machine (VM)

A **Virtual Machine (VM)** is software that allows you to run another Operating System (OS) inside your computer.

Supported Operating Systems:

- Windows
 - Linux
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3. Install MySQL

- Install **MySQL** on your system to create and manage relational databases.
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4. Database as a Service (DBaaS)

Database as a Service (DBaaS) means the database is hosted on the cloud. You don't need to install or manage the database server yourself.

Examples:

- **MongoDB → MongoDB Atlas**
 - **MySQL → FreeMySQLDB**
 - **MySQL → Clever Cloud**
 - **Firebase**
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MongoDB vs MySQL/PostgreSQL

MongoDB

Unstructured Database

Data is stored in **Collections**

Collections contain **Documents**

Documents store data as **Key : Value** pairs

MySQL / PostgreSQL

Structured Database

Data is stored in **Tables**

Tables contain **Rows**

Tables contain **Columns**

MongoDB Structure

Database

↓

Collection

↓

Document

↓

Key : Value

Example

```
{  
  "name": "Sindhu",  
  "age": 21,  
  "city": "Gudivada"  
}
```

MySQL / PostgreSQL Structure

Database

↓

Table

↓

Rows



Columns

Example

	ID	Name	Age	City
1		Sindhu	21	Gudivada